Data analysis and power analysis using the developed package RIQITE

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Abstract

This file illustrates using the RIQITE package by implementing the analyses shown in the main paper.

1 Install the RIQITE package

We first install the package from Github. The github page of the package (https://github.com/li-xinran/RIQITE) contains main functions with detailed explanation in R documentation, as well as a simple illustrating example. Below we provide the code used to analyze the data sets in the paper.

Install via `devtools`, and then load:

```
devtools::install_github("li-xinran/RIQITE")
library(RIQITE)
```

2 Testing monotonicity of an instrumental variable

We get the data from Black et al. (2011), in which Table 1 cross-tabulates month of birth and school entry age (early/on-time/late) in terms of proportions and Table 3 reports the total number of subjects born in January or December (the "discontinuity subsample"): 104,023. We round the total number of subjects to 104,000, and, similar to Fiorini and Stevens (2014), we assume there are equal numbers of subjects born in January and December. Based on these summary statistics, we generate the individual-level data as follows:

```
x <- matrix(0, 2, 3, dimnames=list(Month=c("Dec", "Jan"),
                                   Entry=c("Early", "On-Time", "Late")))
n <- 104000
x["Dec", "On-Time"] <- .85
x["Dec", "Late"] <- .15
x["Jan", "Early"] <- .10
x["Jan", "On-Time"] <- .90
df <- data.frame(BirthMonth = rep(c("Dec", "Jan"), each=n/2),
                 EntryTiming = c(rep("On-Time", n/2 * .85), rep("Late", n/2 * .15),
                                 rep("Early", n/2 * .10), rep("On-Time", n/2 * .90)))
df <- df %>% mutate(EntryAge = NA,
        EntryAge = ifelse(BirthMonth == "Dec" & EntryTiming == "On-Time", 6.75, EntryAge),
        EntryAge = ifelse(BirthMonth == "Dec" & EntryTiming == "Late", 7.75, EntryAge),
        EntryAge = ifelse(BirthMonth == "Jan" & EntryTiming == "On-Time", 23/3, EntryAge),
        EntryAge = ifelse(BirthMonth == "Jan" & EntryTiming == "Early", 20/3, EntryAge))
data <- df[sample(c(1:n)), ] # randomly permute the ordering of units
table(data$BirthMonth, data$EntryTiming)
```

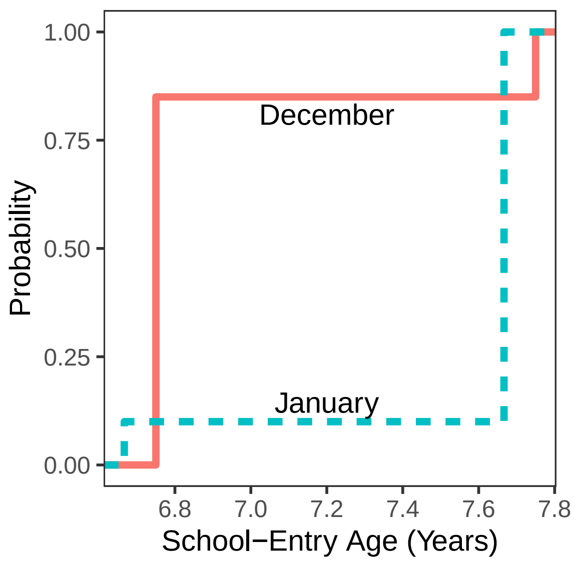
```
##
##       Early  Late On-Time
##   Dec     0  7800   44200
```

```
##    Jan  5200     0   46800
```

The following figure shows the empirical distribution functions of the school-entry age of December-born and January-born subjects. From the figure, most children started school at an older age if they were born in January rather than December.

```
ggplot(data, aes(EntryAge, color = BirthMonth, linetype=BirthMonth)) +
  stat_ecdf(geom="step", show.legend=FALSE, size=1.3) +
  annotate("text", x=c(7.2, 7.2), y=c(.145, .81),
           label=c("January", "December")) +
  scale_x_continuous(breaks=seq(6, 8, .2)) +
  theme_bw() +
  theme( plot.background = element_blank(),
         panel.grid.major = element_blank(),
         panel.grid.minor = element_blank(),
         plot.title=element_text(hjust=.5 ) ) +
  labs(x="School-Entry Age (Years)", y="Probability")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



We first perform the standard regression analysis assessing the relation between the instrument (birthday) and the school-entry age. The first-stage relationship is incredibly strong, with an average effect of -0.667 years and an F statistic of 106,256. This would conventionally be considered persuasive evidence of a valid instrument.

```
lm.fit <- lm(EntryAge ~ (BirthMonth=="Dec"), data)
cat(paste(
  "esimated average effect is", round( as.numeric(lm.fit$coefficients[2]), digits = 3),
  "and F statistic is", round( as.numeric( summary(lm.fit)$fstatistic[1] ), digits = 0),
  "\n"
  ))
```

```
## esimated average effect is -0.667 and F statistic is 106256
```

We then perform the randomization test for the bounded null $H_{\leq 0}$ on the monotonicity of the instrument.

Specifically, we consider three randomization tests using the difference-in-means statistic, Wilcoxon rank sum statistic and Stephenson rank sum statistic with the parameter $s = 10$, respectively. We also conduct the classical Student's t-test. These produce Table 1 in the paper.

```
data$Z = as.numeric( data$BirthMonth == "Dec" )
data$Y = data$EntryAge
n <- length(data$Z)
m <- sum(data$Z)
nperm <- 10^4 # number of permutations for Monte Carlo approximation

# Generate a sample of permutations
Z.perm <- assign_CRE(n, m, nperm)

# difference-in-means
pval.dim <- pval_bound(Z=data$Z, Y=data$Y, c=0, method.list=list(name = "DIM"),
                     alternative = "greater", Z.perm=Z.perm, impute="control")
ci.dim <- ci_bound(Z=data$Z, Y=data$Y, alternative="greater", method.list=list(name="DIM"),
                        Z.perm=Z.perm, impute="control", alpha=0.10)

# Wilcoxon rank sum
pval.wilc <- pval_quantile(Z=data$Z, Y=data$Y, k=n, c=0, alternative = "greater",
                           method.list=list(name="Stephenson", s=2), Z.perm=Z.perm )
ci.wilc <- ci_quantile(Z=data$Z, Y=data$Y, k.vec = n, alternative = "greater",
                method.list=list(name="Stephenson", s=2), Z.perm=Z.perm,  alpha=0.10)

# Stephenson rank sum
pval.steph10 <- pval_quantile(Z=data$Z, Y=data$Y, k=n, c=0, alternative = "greater",
                              method.list=list(name="Stephenson", s=10), Z.perm=Z.perm )
ci.steph10 <- ci_quantile(Z=data$Z, Y=data$Y, k.vec = n, alternative = "greater",
              method.list=list(name="Stephenson", s=10), Z.perm=Z.perm,  alpha=0.10)

# classical Student's t-test
ttest = t.test(data$Y[data$Z==1], data$Y[data$Z==0], alternative = "greater")
pval.ttest = ttest$p.value
ci.ttest = round( ttest$conf.int[1], digits = 3)

# result for Table 1 in the paper
result = data.frame(test=c("diff-in-means", "Wilcoxon", "Stephenson 10", "t test"),
                    pval=c(pval.dim, pval.wilc, pval.steph10, pval.ttest),
                    lower = c(ci.dim, ci.wilc$lower, ci.steph10$lower, ci.ttest))
knitr::kable( result )
```

test	pval	lower
diff-in-means	1	-0.669
Wilcoxon	1	-0.916
Stephenson 10	0	0.084
t test	1	-0.670

Finally, we infer the effect range. The result shows that the range of the effect of birth month on school-entry age is at least 1 year, indicating significant individual effect heterogeneity.

```
tau.max.lower = ci_quantile(Z=data$Z, Y=data$Y, k.vec = n, alternative = "greater",
              method.list=list(name="Stephenson", s=10), Z.perm=Z.perm,  alpha=0.05)$lower
```

```
tau.min.upper = ci_quantile(Z=data$Z, Y=data$Y, k.vec = 1, alternative = "less",
               method.list=list(name="Stephenson", s=10), Z.perm=Z.perm,  alpha=0.05)$upper
print(paste( "lower confidence limit for effect range is",
             max(0, tau.max.lower - tau.min.upper) ))
```

```
## [1] "lower confidence limit for effect range is 1.001"
```

```
save.image("BirthSchoolAge20230601.RData") # save the analysis result
```

3 Evaluating the effectiveness of professional development

We next evaluate a teacher professional development RCT where treated teachers were given a professional development course on electric circuits. We get the data from Heller et al. (2010). The cleaned data, as described in the paper, is part of the package. We load it and estimate the Average Treatment Effect (ATE) via linear regression:

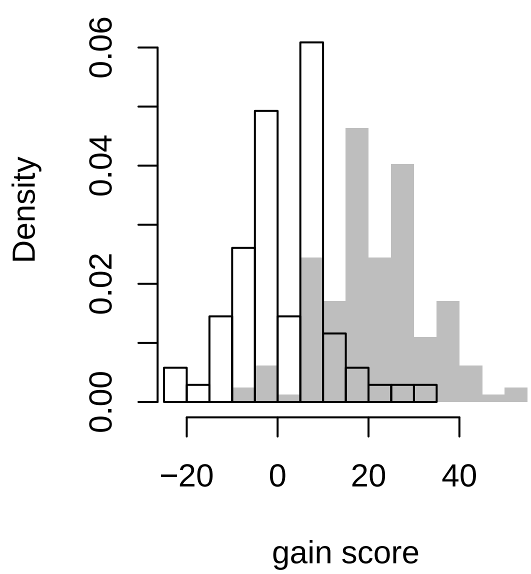
```
data( electric_teachers )

mod <- lm(gain~TxAny, data = electric_teachers)
est_ATE <- round(mod$coefficients[2], digits = 2)
est_ATE
```

```
## TxAny
## 19.77
```

The following figure shows the histograms of the observed gain scores in the treatment and control groups.

```
hist(electric_teachers$gain[electric_teachers$TxAny==1],
     xlim = range(electric_teachers$gain), freq = FALSE, ylim = c(0, 0.065),
     col = "grey", border=FALSE, breaks = 20,
     main = NULL, xlab = "gain score")
hist(electric_teachers$gain[electric_teachers$TxAny==0],
     freq = FALSE, add = TRUE, breaks = 20, col = NULL)
```



We can compare the variation of the units in the treatment and control arms, which can be a clue to how much impact heterogeneity there might be. We do this in effect size units, by first standardizing gain by the control-side variation in outcome at baseline:

```
sd_0 = sd( electric_teachers$Per.1[ electric_teachers$TxAny == 0 ] )
electric_teachers$std_gain = electric_teachers$gain / sd_0
electric_teachers %>% group_by( TxAny ) %>%
    summarise( raw_mean = mean( gain ),
               mean = mean( std_gain ),
               var = var( std_gain ),
               max = max( std_gain ),
               min = min( std_gain ),
               raw_sd = sd( gain ),
               sd = sd( std_gain ) )
```

```
## # A tibble: 2 x 8
##   TxAny raw_mean  mean   var   max    min raw_sd    sd
##   <dbl>    <dbl> <dbl> <dbl> <dbl>  <dbl>  <dbl> <dbl>
## 1     0     2.46 0.189 0.650  2.55 -1.79    10.5 0.806
## 2     1    22.2  1.70  0.726  4.08 -0.510   11.1 0.852
```

If there were no correlation between $Y_i(0)$ and τ_i, the standard deviation of the treatment impacts would be, using $var(Y_i(1)) = var(Y_i(0)) + var(\tau_i)$,

```
tx_het <- sqrt( 0.852^2 - 0.806^2 )
tx_het
```

```
## [1] 0.2761666
```

I.e., around 0.28σ, in effect size units. This would be $0.28 \times sd_0$, or 3.6 in the original scale.

An estimate of the minimum treatment effect possible is the largest control outcome minus the largest treated outcome, which is about $3.16 - -0.63 = 3.79$. The maximum, by contrast, is \$5.06 - -2.22 = 7.28. These are extremely conservative bounds, in that such magnitudes are unlikely to exist in practice.

3.1 Determining an s-value for the test statistic

We plan on using a Stephenson's Rank Sum test, but need to determine an appropriate value of s. We first conduct a sensitivity analysis across a range of scenarios to determine which s have the highest amount of power for the top 10 quantiles, the next 20 quantiles, and the next 40 quantiles.

To illustrate this process, we first run a check on what the best s is for a single scenario. We use the empirical distribution of the control outcomes as our distribution, and calculate a treatment effect (all in standardized units). Our first scenario has impacts be normally distributed and not correlated with $Y_i(0)$, in line with our explorations above.

```
set.seed(303020)
cdat = filter( electric_teachers, TxAny == 0 )
ate = est_ATE / sd_0
R = 1000
nperm = 10^3
s_list = c( 2, 3, 4, 5, 6, 8, 10, 15, 20 )
res <- explore_stephenson_s( s = s_list,
                      n = nrow( electric_teachers ),
                      Y0_distribution = cdat$std_gain,
                      tx_function = "rnorm", ATE = ate,
                      tx_scale = tx_het, rho = 0,
                      nperm = nperm,
                      R = R, calc_ICC = FALSE,
                      targeted_power = FALSE, k.vec = (233-99):233 )
```

We get one row per targeted quantile and explored s, telling us how easy it was to detect effects at that quantile. Here we see how our performance was on the 170th largest effect (of 233 effects).

```
filter( res, k == 170 )
```

```
## # A tibble: 9 x 6
## # Groups:   s [9]
##       s     k power SEpower       q_ci     n
##   <dbl> <dbl> <dbl>   <dbl>      <dbl> <dbl>
## 1     2   170 0       0      -Inf       41.1
## 2     3   170 0       0        -1.13    51.7
## 3     4   170 0.07    0.0119   -0.279   58.1
## 4     5   170 0.352   0.0270   -0.0727  61.6
## 5     6   170 0.543   0.0278    0.00956 63.6
## 6     8   170 0.644   0.0264    0.0589  65.3
## 7    10   170 0.59    0.0293    0.0263  64.3
## 8    15   170 0.349   0.0289   -0.121   57.4
## 9    20   170 0.21    0.0268   -0.260   50.2
```

In our table, we see that s of 6 or 8 gave the highest bound on the impact for this quantile (see the `q_ci` column). The power columns shows the power to reject no treatment effect for the 170th largest effect. Power is very high for $s = 8$, corresponding to the high median confidence bound. The n column is for the entire set of quantiles, and shows the average number of units found as significant: For $s = 8$ we are finding around 63 units as significant across the simulations.

We next make a grid of different scenarios to explore to see if our trends of best s are consistent across different possible scenarios:

```
checks = expand_grid( TxVar = c( 0.25, 0.75 ),
                      tx_dist = c( "rexp", "rnorm" ),
                      rho = c( -0.5, 0, 0.5 ) )

# There is no treatment variation, nor correlation with Y0, when
# the tx impact is a constant.
checks = bind_rows( checks,
                    tibble( TxVar = 0, tx_dist = "constant", rho = 0 ) )
```

We have 13 scenarios. We use `pmap()` to run `explore_stephenson_s` across them. The `explore_stephenson_s()` function will repeatedly generate datasets and analyze them, and then return a summary of the results. It will do this using parallel processing, if we set that flag and if our computer has multiple cores. We specify saving the results for the top 100 quantiles:

```
checks$data = pmap( checks, function( TxVar, tx_dist, rho ) {
    cat( glue::glue("Running {TxVar} {tx_dist} rho={rho}\n" ) )
    explore_stephenson_s( s = s_list,
                          n = nrow( dat ),
                          Y0_distribution = cdat$std_gain,
                          tx_function = tx_dist,
                          ATE = ate, tx_scale = TxVar, rho = rho,
                          nperm = nperm, R = R,
                          calc_ICC = FALSE,
                          parallel = TRUE, n_workers = 5,
                          targeted_power = FALSE, k.vec = (233-99):233 )
} )
checks
s_selector = unnest( checks, cols = "data" )
```

We next aggregate our findings and see what s provides the tightest confidence intervals across a range of quantile groups: the top 10, next 20, next 30, and next 40. For each group, we take the median of the median lower confidence bound, to get a sense of how informative our confidence intervals in each group will tend to be. We plot this vs. s. High points on the curve indicate those s that are more informative:

```
n_units = nrow(electric_teachers)
s_selector = mutate( s_selector,
                     group = cut( k,
                                  breaks = c(0, 233-60, 233-30, 233-10, 233),
                                  labels = c( "v low", "low", "med", "high" ) ) )

s_selector$rho.f = factor( s_selector$rho,
                          levels = c( -0.5, 0, 0.5 ),
                          labels = c( "rho = -0.5", "rho = 0", "rho = 0.5" ) )

avg = s_selector %>%
    group_by( s, group, rho, rho.f ) %>%
    summarise( q_ci = median( q_ci ),
               n = n(), .groups="drop" ) %>%
    filter( group != "v low" )
avg
```

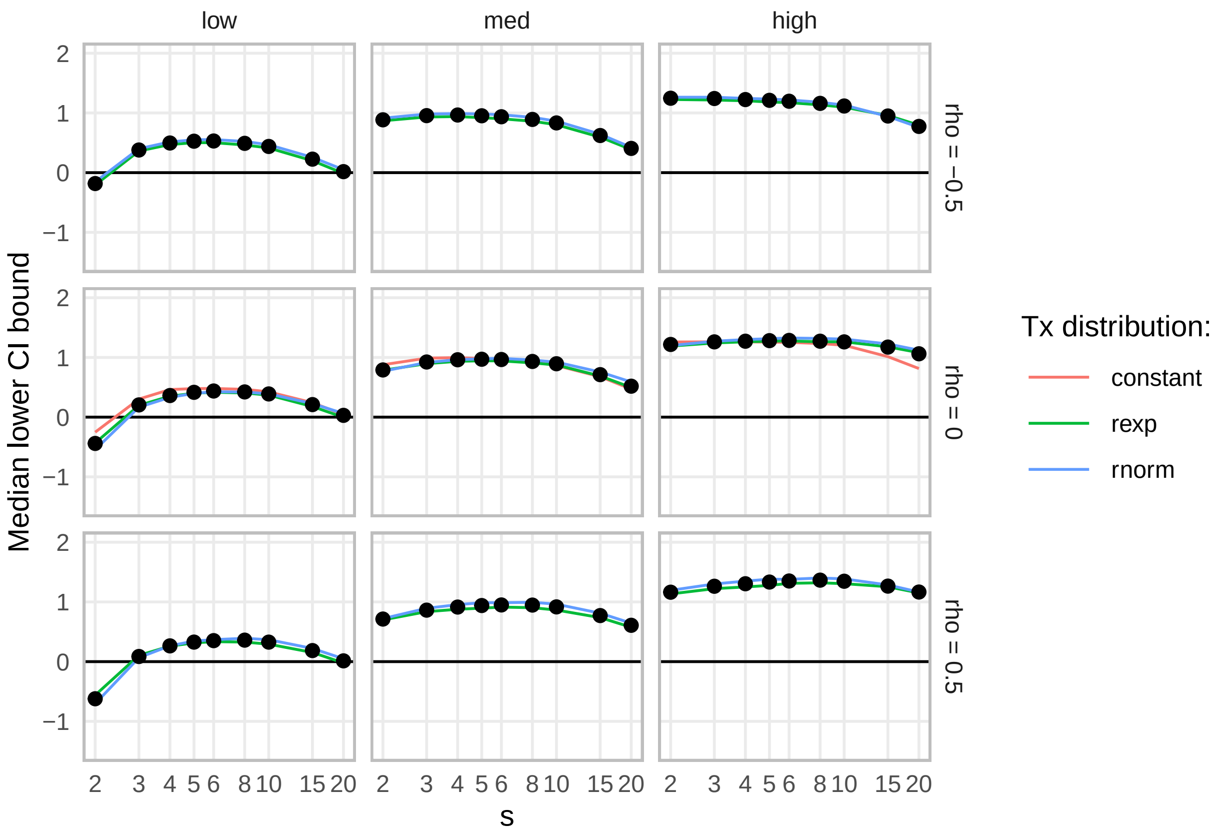
```
## # A tibble: 81 x 6
##        s group   rho rho.f        q_ci     n
##    <dbl> <fct> <dbl> <fct>       <dbl> <int>
##  1     2 low    -0.5 rho = -0.5 -0.183   120
##  2     2 low     0   rho = 0    -0.440   150
##  3     2 low     0.5 rho = 0.5  -0.622   120
##  4     2 med    -0.5 rho = -0.5  0.885    80
##  5     2 med     0   rho = 0     0.790   100
##  6     2 med     0.5 rho = 0.5   0.713    80
##  7     2 high   -0.5 rho = -0.5  1.25     40
##  8     2 high    0   rho = 0     1.22     50
##  9     2 high    0.5 rho = 0.5   1.16     40
## 10     3 low    -0.5 rho = -0.5  0.379   120
## # i 71 more rows
```

```
s_agg = s_selector %>%
    group_by( s, group, rho, rho.f, tx_dist ) %>%
    summarise( q_ci = median( q_ci ) )
```

```
## `summarise()` has grouped output by 's', 'group', 'rho', 'rho.f'. You can
## override using the `.groups` argument.
```

```
s_agg %>%
    filter( group != "v low" ) %>%
    ggplot( aes( s, q_ci ) ) +
    facet_grid( rho.f ~ group ) +
    geom_hline( yintercept = 0 ) +
    #geom_smooth( aes( col=tx_dist ), method = "loess", se = FALSE,
    #             span = 1, lwd= 0.5 ) +
    geom_line( aes( col=tx_dist ), lwd=0.5 ) +
    labs( x = "s", y = "Median lower CI bound", col = "Tx distribution:" ) +
    scale_x_log10( breaks = unique( s_selector$s ) ) +
    theme_minimal() +
    theme( panel.grid.minor = element_blank() ) +
```

```
    geom_point( data = avg, col="black", size=2 ) +
    coord_cartesian( ylim = c( -1.5, 2 ) ) +
    theme( #legend.position="bottom",
           #                 legend.direction="horizontal",
        legend.key.width=unit(1,"cm"),
           panel.border = element_rect(colour = "grey", fill=NA, size=1) )
```



We can also just tally up the winning s across each of the 100 quantiles we are exploring:

```
qis <- s_selector %>%
    group_by( TxVar, tx_dist, rho, group, k ) %>%
    filter( q_ci == max(q_ci) )
table( group = qis$group, s = qis$s, rho = qis$rho )
```

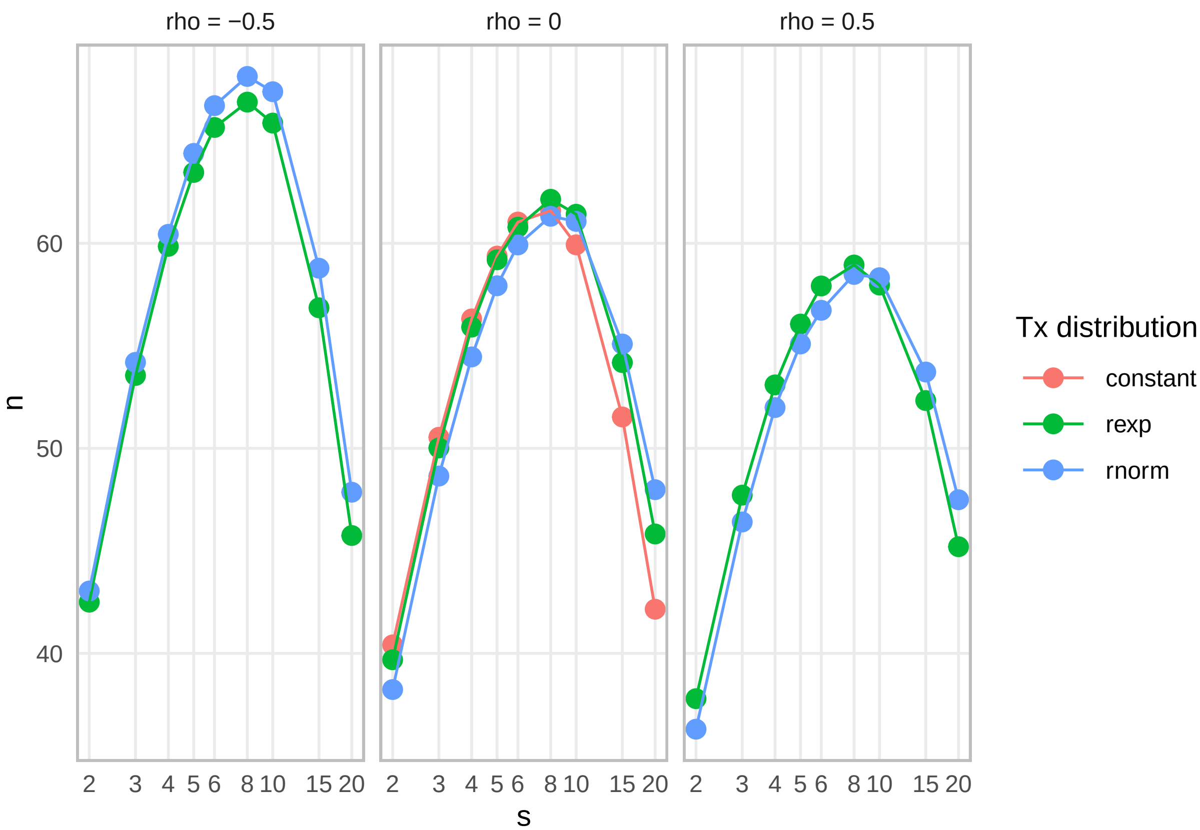
```
## , , rho = -0.5
##
##         s
## group    2  3  4  5  6  8 10 15 20
##   v low  0  0  0  0  2 42 47 39 30
##   low    0  0 13 33 64 10  0  0  0
##   med    0 20 37 23  0  0  0  0  0
##   high  27 13  0  0  0  0  0  0  0
##
## , , rho = 0
##
##         s
## group    2  3  4  5  6  8 10 15 20
##   v low  0  0  0  0  0 32 75 52 41
```

```
##   low     0  0  4  8 83 43 12  0  0
##   med     0  0 26 36 31  7  0  0  0
##   high    0  6 17  0  7  3 14  1  2
##
## , , rho = 0.5
##
##         s
## group    2  3  4  5  6  8 10 15 20
##   v low  0  0  0  0  0 16 59 47 39
##   low    0  0  0  7 34 71  8  0  0
##   med    0  0  0 20 20 27 13  0  0
##   high   0  0  0  7 10  9 10  2  2
```

Each cell in this table records the number of times a particular s value had the highest lower confidence bound for a given scenario and given quantile, divided by the groups of quantiles. We see that for strong confidence intervals on higher quantiles, lower s is good. To obtain more informative findings on the quantiles lower down, higher s is good.

Finally, we can look at the number of units found to be significant across the different scenarios and see which s values identify the most units as significant, on average:

```
n_score <- s_selector %>%
    group_by( rho, rho.f, tx_dist, s ) %>%
    summarize( n = mean( n ), .groups = "drop")
ggplot( n_score, aes( s, n, col=tx_dist ) ) +
    facet_wrap( ~ rho.f, nrow=1 ) +
    geom_point( size=3) + geom_line() +
    labs( col = "Tx distribution:") +
    scale_x_log10( breaks = unique( s_selector$s ) ) +
    theme_minimal() +
    theme( panel.grid.minor = element_blank() ) +
    theme( legend.key.width=unit(1,"cm"),
           panel.border = element_rect(colour = "grey", fill=NA, size=1) )
```



Looking across our results, a wide range of s could be argued for. There is a cost for small s of 2 through 4 for the lower quantiles. The very high s of 10 or above have costs as well, across the simulations: note the down-turn of the power curves in the plot above. Given the apparent shift of the treatment vs. control distributions in our data, either treatment variation is not overly large, or there is a negative correlation between $Y_i(0)$ and τ_i (otherwise the variance of the treated group would be substantially larger than the control group, and they are instead somewhat similar), so we should attend more closely to those simulations that are aligned with these empirical trends. For optimizing the number of units tagged as significant, we typically see $s = 6$ through 10 as performing well across the scenarios. Overall we might adjust our s down a bit to try and optimize multiple goals at once, using, e.g., $s = 6$.

3.2 Analyzing the teacher professional development data

With our selected $s = 6$ in hand, we first test the bounded null hypothesis that all individual effects are non-positive.

```
selected_s = 6
nperm = 10^6
n_units = nrow( electric_teachers )
pval.steph <- pval_quantile(Z=electric_teachers$TxAny, Y=electric_teachers$gain,
                            k=n_units, c=0, alternative = "greater",
                            method.list=list(name="Stephenson", s=selected_s), nperm=nperm )
pval.steph # final p-value
```

```
## [1] 0
```

We then conduct inference for all quantiles of individual treatment effects. The following figures show the 90% lower confidence limits for all quantiles of individual effects using Stephenson rank statistics with s = 6 and s = 2 (i.e., Wilcoxon rank), respectively.

```r
ci6 = ci_quantile( Z=electric_teachers$TxAny, Y=electric_teachers$gain, alternative="greater",
                   method.list=list( name="Stephenson", s=selected_s ), nperm=nperm, alpha=0.10 )
```

We can summarize our results:

```r
summary( ci6, c = c(-0.0001, 0, 6), k = c( 140, 141, 145, 146, 164, 165, 233 ) )
```

```
## CI quantile on 233 units with 164 treated:
## Method: Stephenson-6
## Alternative: greater with alpha=0.1
## Quantiles covered from 1 to 233 (233 quantiles)
## 1e+06 permutations with tolerance of 0.001
##
## Quantile 117 and above (50%, or 117 units) have finite lower CIs
## Finite bounds range from -23.33 to 16.67
## CI for maximal effect is 16.67 and above.
##
## 93 or more units (40%) with effects larger than -1e-04
## ..quantile 141 and above have lower CI bounds > -1e-04
## 88 or more units (38%) with effects larger than 0
## ..quantile 146 and above have lower CI bounds > 0
## 69 or more units (30%) with effects larger than 6
## ..quantile 165 and above have lower CI bounds > 6
##
##
## Table: Selected quantile confidence intervals
##
##    k    lower   upper
## ----  -------  ------
##  140   -0.009     Inf
##  141    0.000     Inf
##  145    0.000     Inf
##  146    0.001     Inf
##  164    3.341     Inf
##  165    6.660     Inf
##  233   16.671     Inf
```

Comparing to Wilcoxon:

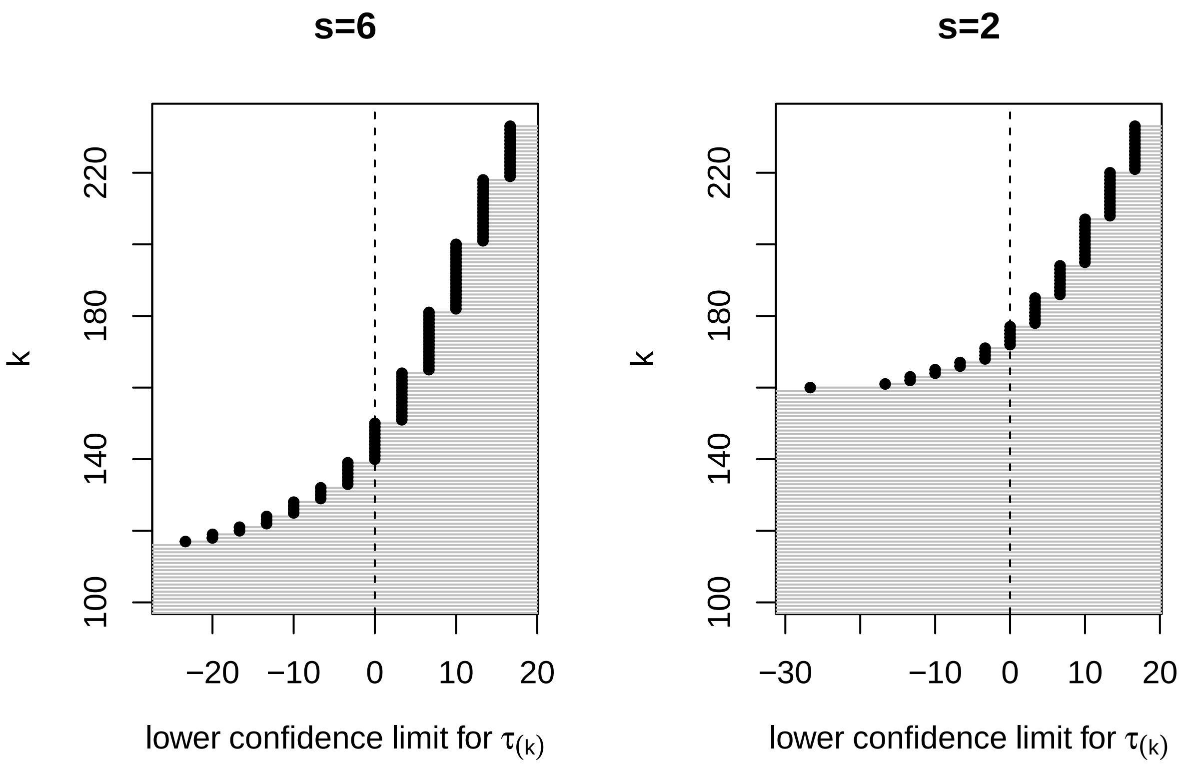
```r
ci2 = ci_quantile( Z=electric_teachers$TxAny, Y=electric_teachers$gain, alternative="greater",
                   method.list=list( name="Stephenson", s=2 ), nperm=nperm, alpha=0.10 )
summary( ci2, c = c( 0, 6 ) )
```

```
## CI quantile on 233 units with 164 treated:
## Method: Stephenson-2
## Alternative: greater with alpha=0.1
## Quantiles covered from 1 to 233 (233 quantiles)
## 1e+06 permutations with tolerance of 0.001
##
## Quantile 160 and above (69%, or 74 units) have finite lower CIs
## Finite bounds range from -26.67 to 16.67
## CI for maximal effect is 16.67 and above.
##
## 59 or more units (25%) with effects larger than 0
## ..quantile 175 and above have lower CI bounds > 0
## 48 or more units (21%) with effects larger than 6
```

```
## ..quantile 186 and above have lower CI bounds > 6
```

We can also visualize the bounds:

```r
par(mfrow = c(1, 2))
s_title = glue::glue( "s={selected_s}" )
plot_quantile_CIs(ci6, k_start = 102, main = s_title )
plot_quantile_CIs(ci2, k_start = 102, main = "s=2" )
```



Finally, we infer the effect range by conducting two tests, one for greater and one for less. We then can say that there are effects at least as small as some value and at least as large as another.

```r
tau.max.lower = ci_quantile( Z=electric_teachers$TxAny, Y=electric_teachers$gain, k.vec=n_units,
                             alternative="greater", method.list=list( name="Stephenson", s=selected_s ),
                             nperm=nperm, alpha=0.05 )$lower
tau.min.upper = ci_quantile( Z=electric_teachers$TxAny, Y=electric_teachers$gain, k.vec=1,
                             alternative="less", method.list=list( name="Stephenson", s=selected_s ),
                             nperm=nperm, alpha=0.05 )$upper
max(0, tau.max.lower - tau.min.upper)
```

```
## [1] 0
```

In this case, we have no evidence of effect heterogeneity (all the effects could be the same).

4 Evaluating the effects of six-month nutrition therapy

We get the data for the homefood study from https://doi.org/10.7910/DVN/38X3LX. We download the SPSS Binary (Original File Format) there. The details of the study can be found at https://clinicaltrials.gov/ct2/show/NCT03995303.

We focus on the effect of the treatment on the increase of lean body mass (kg) measured before and after the trial. We exclude two units with missing outcomes, resulting in 52 treated units and 52 control units.

```
dat$change = dat$lean_mass_kg_1 - dat$lean_mass_kg_0
paste("There are", sum(is.na(dat$change)), "units with missing outcomes.")
```

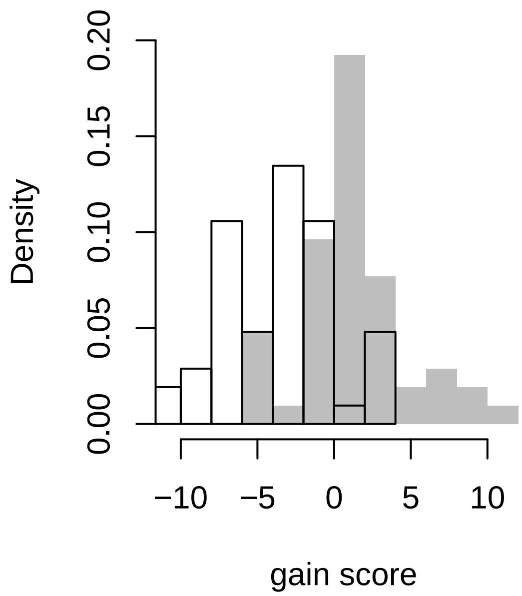
```
## [1] "There are 2 units with missing outcomes."
```

```
dat = dat[ !is.na(dat$change), ] # delete units with missing outcomes
n = nrow(dat)
dat = dat[sample(n), ] # randomly permute the order of units
table(dat$group)
```

```
##
##      control intervention
##           52           52
```

The following figure shows the histograms of the lean body mass changes in treated and control groups, respectively.

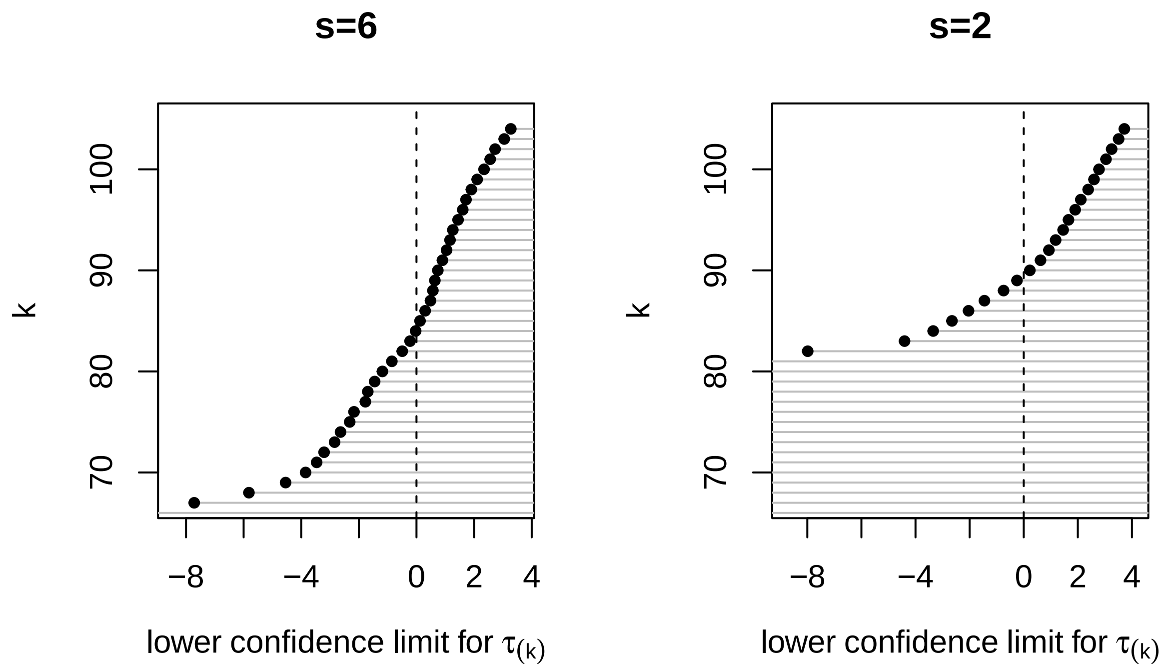
```
hist(dat$change[dat$group=="intervention"], xlim=range(dat$change), freq=FALSE,
     ylim=c(0,0.20), col="grey", border=FALSE, breaks = 10, main=NULL, xlab="gain score")
hist(dat$change[dat$group=="control"], freq=FALSE, add=TRUE, breaks=10, col=NULL)
```



We infer the lower confidence limits for all quantiles of individual treatment effects. The following two figures show the 90% lower confidence limits for all quantiles of individual effects using Stephenson rank statistics with s = 6 and s = 2 (i.e., Wilcoxon rank), respectively

```
nperm = 10^6
ci6 = ci_quantile( Z=as.numeric(dat$group == "intervention"), Y=dat$change,
                   alternative="greater", method.list=list(name="Stephenson", s=6),
                   nperm=nperm, alpha=0.10 )
ci2 = ci_quantile( Z=as.numeric(dat$group == "intervention"), Y=dat$change,
                   alternative="greater", method.list=list(name="Stephenson", s=2),
                   nperm=nperm, alpha=0.10 )

par(mfrow = c(1, 2))
plot_quantile_CIs(ci6, main="s=6", k_start = n-37)
plot_quantile_CIs(ci2, main="s=2", k_start = n-37)
```



References

Black, S. E., Devereaux, P. J., and Salvanes, K. G. (2011). Too young to leave the nest? the effects of school starting age. *Review of Economics and Statistics*, 93:455–467.

Fiorini, M. and Stevens, K. (2014). Assessing the Monotonicity Assumption in IV and Fuzzy RD designs. Working papers, University of Sydney, School of Economics.

Heller, J. L., Shinohara, M., Miratrix, L., Hesketh, S. R., and Daehler, K. R. (2010). Learning science for teaching: Effects of professional development on elementary teachers, classrooms, and students. *Proceedings from Society for Research on Educational Effectiveness.*